

Software License & Subscription Management System



Design & Implementation of a Relational Database for Centralized License Governance

STUDENT Mihir Kalway	DEPARTMENT SOT	PROFESSOR Dr. Kiran Mayee Adavala	DATE 17th Sept 2025
-------------------------	-------------------	--------------------------------------	------------------------

Licenses expire silently. Seats get over-allocated.



THE PROBLEM

Organizations run many licensed products across departments with no centralized control — licenses expire unnoticed or seats are allocated beyond what was purchased.



SCOPE

In scope — vendors, products, license types, purchases, allocation, subscriptions, renewals, compliance, reporting.

Out of scope — deployment, payment gateways, real-time telemetry.



OBJECTIVES

Normalized 3NF schema with strong integrity constraints. Prevent over-allocation and duplicate keys. Enable expiry, utilization, and compliance reporting. Java/Python front-end for CRUD.

User Roles & Functional Requirements



IT Administrator

Maintains inventory, allocates and reclaims licenses



Department Manager

Requests licenses for team members



Procurement Manager

Records purchases and manages vendor renewals



Compliance Auditor

Runs compliance checks and audit reports



End User

Uses allocated software on their device

FUNCTIONAL REQUIREMENTS

- ✓ Record vendors, products, purchases with unique license keys
- ✓ Allocate licenses within purchased seat quantity
- ✓ Track subscriptions and process renewals
- ✓ Reclaim licenses when users change roles
- ✓ Run compliance checks against active allocations
- ✓ Generate expiry, utilization, and compliance reports

12 core entities

forming the complete data model

• Vendor

Supplier of software products

• Software Product

Licensed product catalog

• License Type

Perpetual / Sub / Trial / OEM

• Purchase

Purchase order with quantity

• License

Individual license with key

• Subscription

Recurring billing record

• Department

Organizational business unit

• Users

License recipients

• Device

Hardware holding a license

• Allocation

License → user/device link

• Renewal

Extends license validity

• Compliance Check

Periodic audit log

Structured Entity-Relationship Model

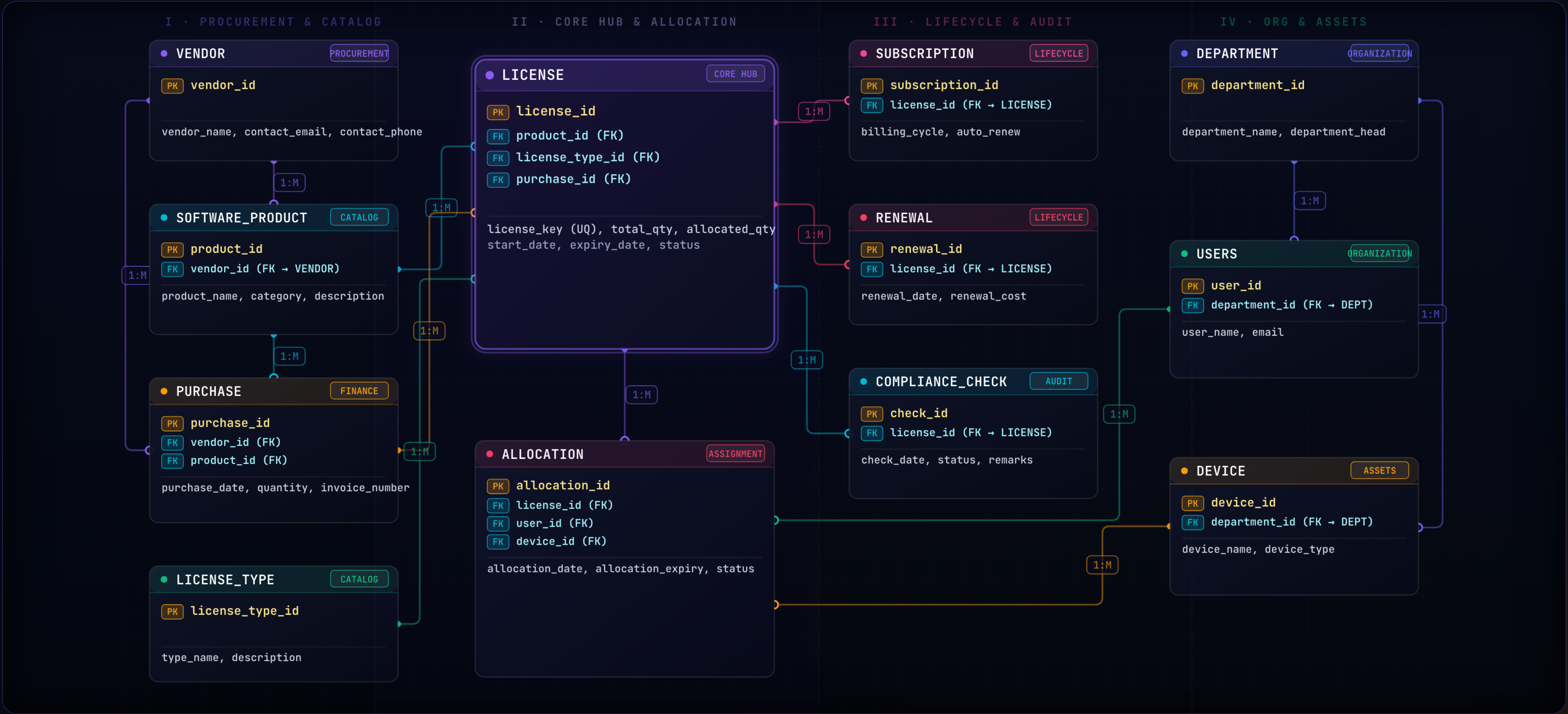
All Tables (12)

Procurement & Catalog (4)

Core Hub & Allocation (2)

Lifecycle & Audit (3)

Organization & Assets (3)



Architectural Legend · PK Primary Key · FK Foreign Key · 1:M One-to-Many Relational Cardinality

12 ENTITIES · 14 RELATIONSHIPS · 3NF

Cardinalities & Meanings

SOURCE	TARGET	CARDINALITY	MEANING
Vendor	Product	1 : M	One vendor supplies many software products
Product / License Type / Purchase	License	1 : M	Each entity generates multiple license records
License	Subscription	1 : M opt	Recurring billing for subscription-type licenses
License	Allocation	1 : M	One license (N seats) allocated to many users or devices
License	Renewal	1 : M	A license can be renewed multiple times over its lifetime
License	Compliance Check	1 : M	Compliance checks run repeatedly per license
Department	Users / Device	1 : M	A department owns many users and devices

Relational Schema, Part 1

• VENDOR		
vendor_id	INT	PK
vendor_name	VARCHAR(100)	NOT NULL
contact_email	VARCHAR(150)	UNIQUE
contact_phone	VARCHAR(20)	
address	TEXT	

• LICENSE_TYPE		
license_type_id	INT	PK
type_name	VARCHAR(50)	NOT NULL
description	TEXT	

• SOFTWARE_PRODUCT		
product_id	INT	PK
product_name	VARCHAR(150)	NOT NULL
vendor_id	INT	FK → VENDOR
category	VARCHAR(50)	
description	TEXT	

• PURCHASE		
purchase_id	INT	PK
vendor_id	INT	FK → VENDOR
product_id	INT	FK → PRODUCT
purchase_date	DATE	NOT NULL
quantity	INT	CHECK > 0
invoice_number	VARCHAR(50)	UNIQUE

PK Primary key FK Foreign key UNIQUE Unique constraint

Relational Schema, Part 2

● SUBSCRIPTION			● DEPARTMENT			● USERS			● DEVICE		
subscription_id	INT	PK	department_id	INT	PK	user_id	INT	PK	device_id	INT	PK
license_id	INT	FK → LICENSE	department_name	VARCHAR(100)	UNIQUE NOT NULL	user_name	VARCHAR(100)	NOT NULL	device_name	VARCHAR(100)	NOT NULL
billing_cycle	VARCHAR(20)		department_head	VARCHAR(100)		email	VARCHAR(150)	UNIQUE NOT NULL	device_type	VARCHAR(50)	
auto_renew	BOOLEAN	DEFAULT FALSE				department_id	INT	FK → DEPT	department_id	INT	FK → DEPT

The database enforces these rules

Not the application layer — the schema itself.

● Unique License Keys	UNIQUE ON license.license_key
● Allocation Within Seat Count	CHECK (allocated_qty ≤ total_qty)
● Valid Allocation Period	CHECK (expiry ≥ allocated_date)
● No Expired License Assignment	WHERE status = 'Active'
● Positive Renewal Cost	CHECK (renewal_cost > 0)
● Referential Integrity	FOREIGN KEY ... REFERENCES

↑ Click any rule to expand details